INDIAN ECONOMY

LEVEL II





LECTURES

AN OVERVIEW OF THE TELECOM SECTOR SINCE ITS PRIVATISATION

MODULE - 45

A GLANCE AT THE LIBERALISATION OF TELECOM SECTOR

BSNL

1994

- **Privatisation of telecom sector**
- **Huge upfront Fixed License Fee**

1999

Revenue Sharing Mechanism in place of Fixed License Fee

- The telecom sector was liberalized under the National Telecom Policy, 1994.
- Various licences were issued to companies under Section 4 of the Indian Telegraph Act, 1885.
- However, as the fixed licence fee was very high and the telecom service providers consistently defaulted in making the payments, the National Telecom Policy of 1999 gave an option to the licencees to migrate from a fixed licence fee to a revenue-sharing mechanism and was made applicable in the year 1999.

QUARTER CENTURY OF TELECOM PRIVATISATION ... VARIOUS POLICY STANCES AND DISRUPTIONS

1990s

Way back in 1990s, some companies bid exorbitant amounts in their greed to garner spectrum and later approached the then Govt. to convert the upfront fees to a revenue share under the New Telecom Policy, 1999.

EARLY 2000s

- In 2001, fixed-line telephony operators were allowed into the mobile sector through the back door.
- Subsequently, the 2G spectrum scam broke out.
- Due to various reasons, around 10 mobile operators had to shut down operations, after investing billions of dollars.

IN RECENT YEARS

- Entry of Reliance Jio created a huge disruption as it unleashed 4G services, with free voice calls and data services priced at a massive discount.
- Questions have been raised about the regulatory actions.

Amid these chaos, multiple litigations were fought between operators and Government over things like, what constitutes AGR for paying licence fees and SUC and whether Interconnect Usage Charges should be zero or not!

GROSS REVENUE IN TELECOM SECTOR

- Installation charges
- Sale proceeds of handsets (or any other equipment)
- Interest
- Dividend
- Value-added services
- Access or interconnection charges
- Roaming charges
- Revenue from infrastructure sharing

For the purpose of arriving at the "Adjusted Gross Revenue (AGR)", charges like PSTN/PLMN related call charges, service tax etc., shall be excluded from the Gross Revenue.

THE SUPREME COURT JUDGEMENT FINALLY SETTLED THE DUST BETWEEN THE OPERATORS AND THE GOVERNMENT ABOUT AGR

TELECOM OPERATORS' VIEW

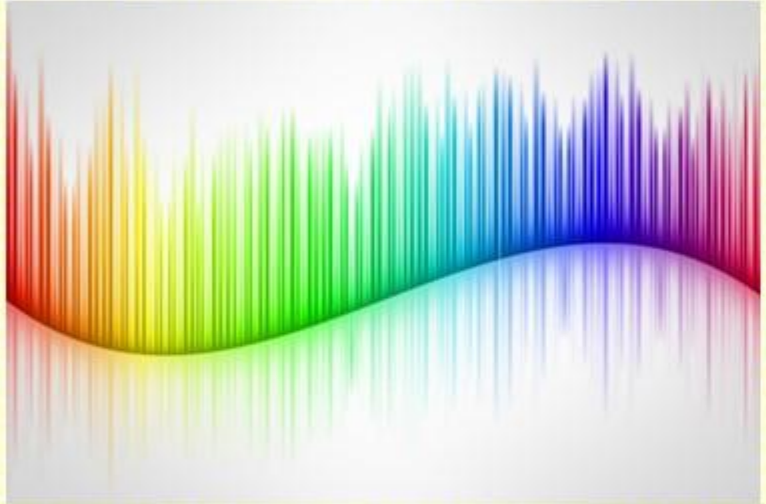
As per the operators, the Department of Telecom changed the definition of AGR in 2002-03 to include non-telecom income, such as foreign exchange gains and interest earned from bank deposits.

INFERENCE BY THE SUPREME COURT

- The Supreme Court relied on the "Migration Agreement" offered to the Telecom operators in 1999, where the Centre converted the upfront money committed by the mobile players as a licence fee towards annual revenue share.
- Supreme Court held that, the DoT was clear from day one, on what would constitute AGR.
- Therefore, it was unfair on the part of the operators to dispute the definition, after having taken benefit of the migration package. Finally, the Supreme Court gave judgement in favour of the Government.

FIVE MAJOR REASONS FOR POOR HEALTH OF THE TELECOM PLAYERS

1 High cost of spectrum



2 Mismanagement and delay in rolling out 4G and poor service quality, till the entry of Reliance Jio

3 Competition among the players, especially allegations of predatory pricing on Reliance Jio

4 High licence fee and Spectrum Usage Charges

5 High cost of capital in India due to higher interest rates

POOR HEALTH OF THE TELECOM WAS UNDERSTOOD BY THE GOVERNMENT!

STRUCTURAL REFORMS ANNOUNCED IN 2022 IN THE TELECOM SECTOR

- Through Rationalization of Adjusted Gross Revenue (AGR), non-telecom revenue will be excluded on prospective basis from the definition of AGR.
- Bank Guarantees (BGs) against Licence Fee (LF) have been rationalized.
- Delayed payments of Licence Fee (LF)/Spectrum Usage Charge (SUC) will attract lesser interest rate now.
- In future auctions, tenure of spectrum increased from 20 to 30 years.
- No Spectrum Usage Charge (SUC) for spectrum acquired in future spectrum auctions.
- 100 percent Foreign Direct Investment (FDI) under automatic route has been permitted in Telecom Sector .

WHAT ARE ELECTROMAGNETIC WAVES?

- Electromagnetic waves are formed, when an electric field comes in contact with a magnetic field.
- Examples of EM waves are radio waves, microwaves, infrared waves, X-rays, gamma rays, etc.

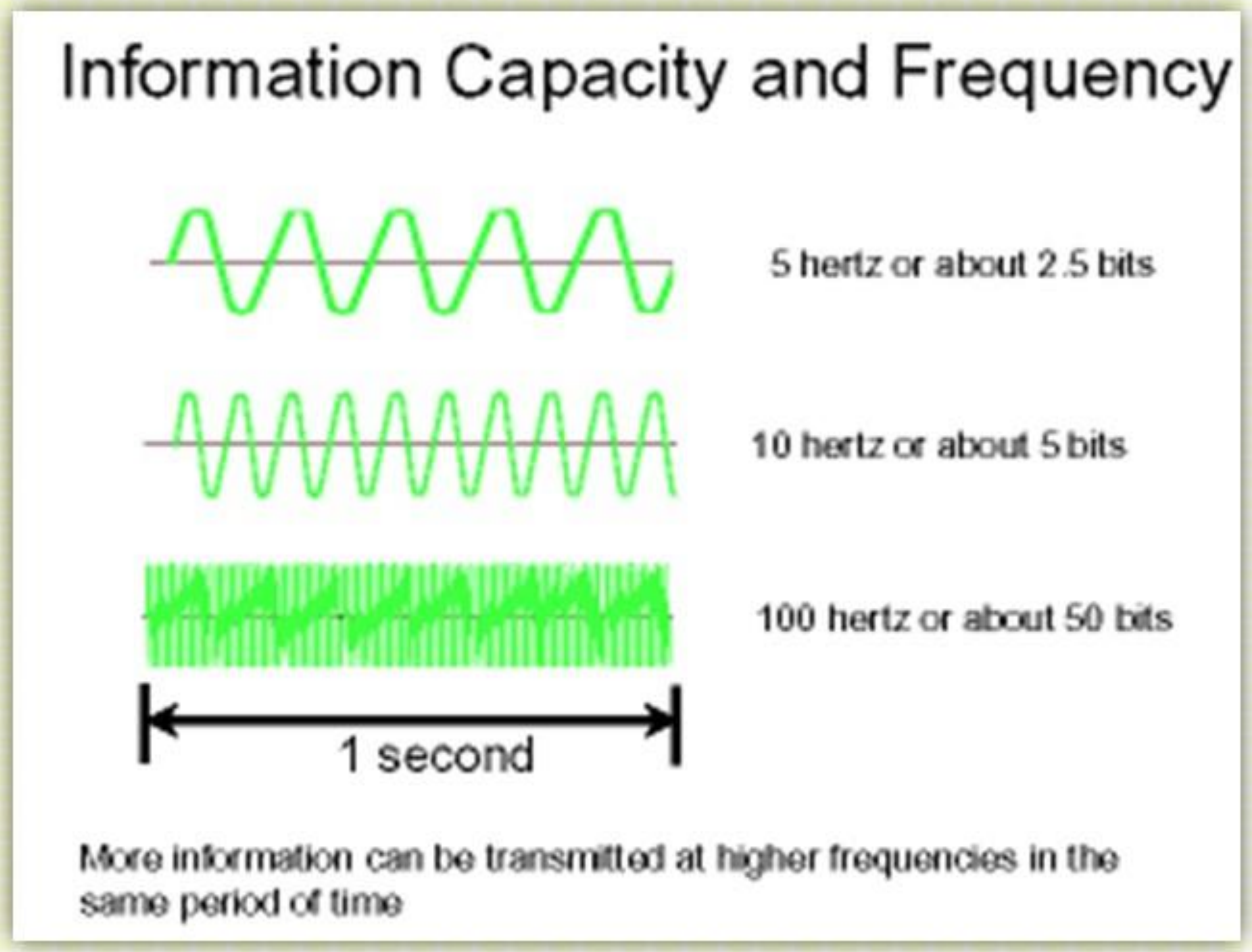
WHAT IS SPECTRUM?

- Spectrum stands for the portion of the electromagnetic wave range that is suitable for communication purposes.
- It is a huge economic resource.
- It also provides unimaginable benefits to any population and it is controlled by the Government.

EM WAVES ARE SPLIT INTO A RANGE OF FREQUENCIES!

- EM waves can be split into a range of frequencies. The set of all possible frequencies is called the electromagnetic spectrum.
- EM waves propagate through space at different radio frequencies.
- In a nut shell, we can say, spectrum auctions relate to the radio frequencies allocated to the mobile industry and other sectors for communication over the airwaves.
- The use of the radio spectrum is regulated, access is controlled and rules for its use enforced because of the possibilities of interference between uncoordinated uses.

HOW DO YOU DIFFERENTIATE ONE SLICE FROM THE OTHER, SAY 700 MHz FROM 800 MHz?



- Hertz is a measure of the number of cycles per second.
- 1 Megahertz means 1 million hertz.
- Spectrum bands have different characteristics, and this makes them suitable for different purposes.
- Low-frequency transmissions can travel greater distances before losing their integrity, and they can pass through dense objects more easily. Less data can be transmitted over these radio waves.
- Higher-frequency transmissions carry more data, but are poorer at penetrating obstacles.

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5G REQUIRES BOTH LOW BAND SPECTRUM & HIGH BAND SPECTRUM!

WHY LOW BAND SPECTRUM?

- Low Band Spectrum is popular for providing wider coverage. However, the speed and latency may not match the requirements for 5G. Speed is subject to the proximity to the source.
- Additionally, signals from the low band spectrum can travel through windows and walls, which is not the case with high band.

The mid band spectrum falls somewhere between the two. It can carry sizable data over longer distances and maintain increased speeds.

WHY HIGH BAND SPECTRUM?

- Example for high band spectrum is 26 GHz. 1 GHz equals 1,000 MHz.
- The high band spectrum can provide speeds of up to 2Gbps, but is unable to travel longer distances, at times, less than a mile.

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ROLLOUT OF 5G ... HUGE TRANSFORMATION ON THE CARDS!

1 EDUCATION AND HEALTHCARE

PM e-Vidya aims to unify all efforts related to digital/online/on-air education to enable equitable multi-mode access to education.

- Full potential of digital education can be unleashed. Expanding on PM e-Vidya - 5G can deliver high quality educational content through mobile applications to every student in the country. 5G will also provide a major impetus to digital universities.
- In healthcare, the Ultra-Reliable Low Latency Communication (URLLC) will enable user friendly point of care diagnostics and creation of much needed connected ambulances.
- A hospital run private 5G network will enable even a handful of doctors and nursing staff to provide quality care simultaneously to several people.

ROLLOUT OF 5G ... HUGE TRANSFORMATION ON THE CARDS!

2 NEXTGEN BANKING

- With the help of geospatial information systems, we can reach the next level of simple, seamless and secure payments such as one tap payment and cashier less store models.
- Similarly, the payments bank model can be expanded towards completely mobile formal banking system. This enables the citizens to securely access various bank facilities through a virtual branch experience.

3 TRANSPORTATION

- In transportation and mobility, a network of electric vehicles and charging stations can be created optimizing the availability of the charging infrastructure.
- It enhances the cost effectiveness of EV ecosystem alongside the launch of the "drones as a service" ecosystem in India. The URLLC feature will be crucial for navigation and drone traffic control.
- The deployment of machine vision with software enabled automated guided vehicles can help in better port space management.

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ROLLOUT OF 5G ... HUGE TRANSFORMATION ON THE CARDS!

4 AGRICULTURE AND RENEWABLE ENERGY

- Firms can be equipped with diverse range of sensors to continuously monitor the factors impacting the health of the crops.
- A small farmer with little virtual training can improve irrigation efficiency as well as crop leads through 5G.
- The sensors provided in renewable energy firms can become more effective and efficiency can be radically improved.

5 MANUFACTURING AND INDUSTRY

- 5G private networks will be the cornerstone of industry 4.0.
- These networks connect an array of Internet of Things sensors and devices and automate the scheduling of various processes based on intelligent algorithms.

ROLLOUT OF 5G ... HUGE TRANSFORMATION ON THE CARDS!

6 EFFICIENT SERVICE DELIVERY

- In governance and public safety, service delivery and citizen management efforts can be improved with faster and safer digital identity verification.
- This enable faster implementation of Direct Benefit Transfers and other such schemes.

DBT Bharat

Apna Adhikar, Apne Dwar

7 MONITORING THE PUBLIC PLACES

- Real time automated monitoring of public spaces and traffic using city owned private 5G networks will improve public safety and congestion in India's metro cities.
- It also improves the efficiency of the projects under the Smart Cities Mission.

Smart City

MISSION TRANSFORM-NATION

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